

Assignment-1:

```
abstract class UniversityMember {  
    protected String memberID;  
    protected String name;  
    UniversityMember(String memberID, String name) {  
        this.memberID = memberID;  
        this.name = name;  
    }  
    public void displayInfo() {  
        System.out.println("Name of the member:" + name);  
        System.out.println("Member ID:" + memberID);  
    }  
    abstract void performDuty();  
}  
  
public class Employee extends UniversityMember {  
    protected Stringdouble salary;  
    Employee(String memberID, String name,  
             double salary) {  
        super(memberID, name);  
        this.salary = salary;  
    }  
    public void showSalary() {  
        System.out.println("Salary: " + salary);  
    }  
}
```

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@Override

```
void performDuty() {  
    System.out.println("Performs duties");  
}
```

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```
public class AcademicStaff extends Employee {  
    protected String department;  
    AcademicStaff (String memberID, String name, double  
        salary, String department) {  
        super(memberID, name, salary);  
        this.department = department;  
    }
```

```
public void showDept () {
```

```
    System.out.println(name + "is a teacher of" + department);  
}
```

@Override

```
void performDuty () {
```

```
    System.out.println(name + "teaches Students");  
}
```

```
}
```



```
public class Professor extends AcademicStaff {
```

```
    private String researchArea;
```

```
    private intString numofPublications;
```

```
    private double researchGrant;
```

```
    Professor(String memberID, String name, double double salary, String department, String researchArea, int numofPublications, double researchGrant) {
```

```
        super(memberID, name, salary, department);
```

```
        this.researchArea = researchArea;
```

```
        this.numofPublications = numofPublications;
```

```
        this.researchGrant = researchGrant;
```

```
    }
```

```
    public void ResearchArea() {
```

```
        System.out.println("Research on" + researchArea);
```

```
        System.out.println("Number of publications + numofPublications);
```

```
        System.out.println("Research Grant" + researchGrant);
```

```
    }
```

```
@Override
```

```
    void performDuty() {
```

```
        System.out.println(name + "is teaching and supervising research");
```

```
    }
```

```
}
```

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```

public void AdministrativeOfficers extends Employee{
    private String officeSection;
    AdministrativeOfficers(String memberID, String name,
        double salary, String officeSection){
        super(memberID, name, salary);
        this.officeSection = officeSection;
    }
    public void manageOffice(){
        System.out.println(name + " manages" +
            officeSection);
    }
    @Override
    void performDuty(){
        System.out.println(name + " handles administrative
            tasks");
    }
}

```

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```

import java.util. ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<UniversityMember> member = new ArrayList<>(X);
        member.add(new Professor("101", "Dr. Rahman", 100000,
            "ICT", "Cyber Security", 20, 500000));
        member.add(new AdministrativeOfficer("1201",
            "Mr. Ahmed", 50000, "Student welfare");
        for (int i=0; i<member.size(); i++) {
            UniversityMember mem = new member.get(i);
            mem.DisplayInfo();
            mem.performDuty();
            if (mem instanceof Employee) {
                Employee e = (Employee) mem; e.showSalary();
            }
            if (mem instanceof AcademicStaff) {
                AcademicStaff a = (AcademicStaff) mem;
                a.showDept();
            }
            if (mem instanceof Professor) {
                Administrative Professor p = (Professor) mem;
                p.ResearchArea();
            }
            if (mem instanceof AdministrativeOfficer) {
                AdministrativeOfficer ao = (AdministrativeOfficer) mem;
                ao.manageOffice();
            }
        }
    }
}

```

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